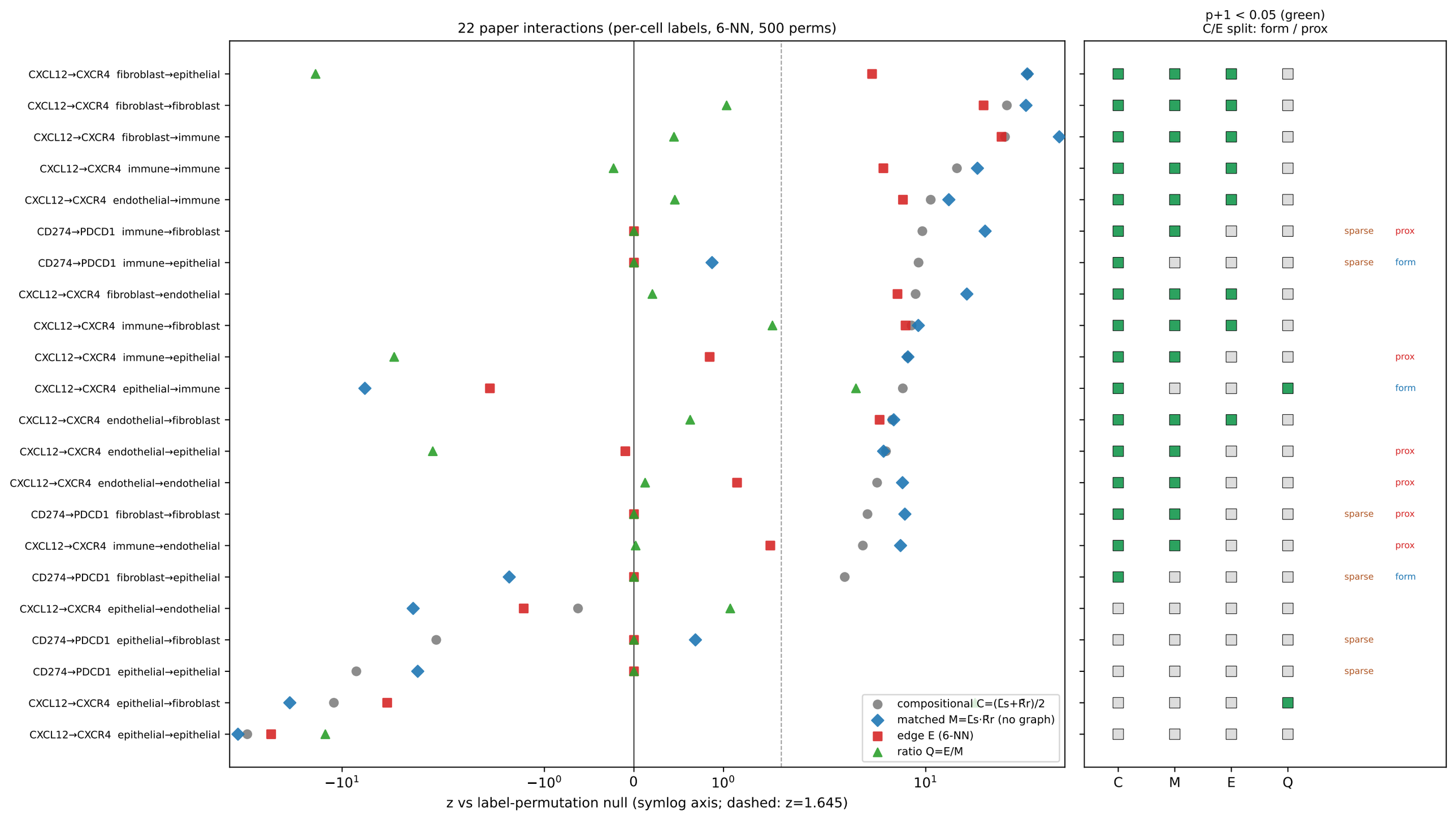
22 paper interactions (per-cell labels, 6-NN, 500 perms)
p+1 < 0.05 (green)
C/E split: form / prox
CXCL12→CXCR4 fibroblast→epithelial
CXCL12→CXCR4 fibroblast→fibroblast
CXCL12→CXCR4 fibroblast→immune
CXCL12→CXCR4 immune→immune
CXCL12→CXCR4 endothelial→immune
CD274→PDCD1 immune→fibroblast
CD274→PDCD1 immune→epithelial
CXCL12→CXCR4 fibroblast→endothelial
CXCL12→CXCR4 immune→fibroblast
CXCL12→CXCR4 immune→epithelial
CXCL12→CXCR4 epithelial→immune
CXCL12→CXCR4 endothelial→fibroblast
CXCL12→CXCR4 endothelial→epithelial
CXCL12→CXCR4 endothelial→endothelial
CD274→PDCD1 fibroblast→fibroblast
CXCL12→CXCR4 immune→endothelial
CD274→PDCD1 fibroblast→epithelial
CXCL12→CXCR4 epithelial→endothelial
CD274→PDCD1 epithelial→fibroblast
CD274→PDCD1 epithelial→epithelial
CXCL12→CXCR4 epithelial→fibroblast
CXCL12→CXCR4 epithelial→epithelial
sparse prox
sparse form
prox
form
prox
prox
sparse prox
prox
sparse form
sparse
sparse
compositional C=(Ls+Rr)/2
matched M=Ls·Rr (no graph)
edge E (6-NN)
ratio Q=E/M
−10^1
−10^0
0
10^0
10^1
C
M
E
Q
z vs label-permutation null (symlog axis; dashed: z=1.645)